Sandun Munasinghe

Machine Learning & AI | LLM Research | Deep Learning | Python & PyTorch

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M.S. Computer Science candidate (Machine Learning specialization) at Stevens Institute of Technology, specializing in Large Language Models (LLMs), Natural Language Processing (NLP), and Deep Learning. Pretraining a GPT-style transformer from scratch in PyTorch, building LLM evaluation and interpretability pipelines as a graduate research assistant, and developing multi-agent Generative AI systems. Brings a Chemical Engineering background and 5+ years of industry experience applying data-driven analysis to complex technical problems.

TECHNICAL SKILLS

Programming & Data Engineering: Python | SQL | NumPy | Pandas | SciPy | Apache Spark | ETL Pipelines

Machine Learning & Deep Learning: PyTorch | TensorFlow | scikit-learn | Neural Networks | Statistical Modeling

Generative AI & LLMs: Transformers | Hugging Face | Pretraining | RAG | LangChain | LangGraph | Multi-Agent Systems | OpenAI API | Prompt Engineering | Open-Weight Models(Gemma, Llama)

LLM Evaluation & Interpretability: TruthfulQA | ROUGE | Logit Lens | Adversarial Prompting | Sentence-Transformers

Software Engineering & Cloud: FastAPI | Async Python | Docker | Git | CI/CD | AWS

Machine Learning & AI Research Experience

Graduate Research Assistant at Stevens Institute of Technology (Advisor: Zining Zhu) *Jan 2026 - Present*
Stevens Institute of Technology | *ECAI Lab*
LLM Truthfulness and Interpretability Research (GitHub)
Python • PyTorch • Hugging Face • LLM Evaluation • Model Interpretability

- Developed an automated LLM evaluation pipeline using 790 TruthfulQA questions across four prompting conditions to investigate adversarial effects on model truthfulness.
- Evaluated GPT-5.6 Sol and Gemma 3 4B IT using automated truthfulness assessment and ROUGE metrics to analyze changes in model behavior.
- Developed a logit-lens analysis framework to investigate layer-wise model representations and changes in token-level predictions under adversarial prompting.

TECHNICAL PROJECTS

Multi-Agent LLM Conflict Resolution (GitHub) | NLP Project, Stevens Institute of Technology *Spring 2026*
Python • LangGraph • NLP • FastAPI • Next.js

- Built 4 LangGraph orchestration strategies (majority voting, structured debate, hierarchical authority, evidence-weighted consensus) with sentence-transformer deadlock detection and judge-mediated resolution
- Streamed multi-agent debates in real time via SSE on an async FastAPI backend with a Next.js frontend and batch experiment runner
- Tracked 9 behavioral metrics across 32 scenarios; found misinformation agents outperformed factual agents in 88% of cases (p=0.006)

Chemical Engineering Domain Language Model (GitHub) | *ML and AI Project, Stevens Institute of Technology* *Fall 2026*
Python • PyTorch • Transformers • NLP • Data Engineering

- Building a GPT-2–style decoder-only transformer from scratch in PyTorch, to be pretrained on a chemical engineering corpus of research papers and technical references
- Engineered automated pipelines to collect 3,000+ arXiv LaTeX source archives and Wikipedia articles, with text extraction, cleaning, quality scoring, and domain-relevance filtering
- Processed 2,300+ research papers into a curated 10M-token training corpus
- Next phase: extending the model into an AI troubleshooting assistant that diagnoses equipment faults, identifies root causes, and recommends corrective actions, supported by deep learning process models for optimization

ADDITIONAL PROFESSIONAL EXPERIENCE

Operations & Customer Service Associate Manager | United States Postal Service Oct 2018 – Oct 2023

- Applied demand variability analysis to redesign staffing models across a 200K+ daily package operation, reducing overtime costs by 20%
- Led deployment of automated sorting systems (SDUS): troubleshot integration issues and trained 20+ staff, improving system adoption
- Communicated performance insights and operational decisions to non-technical stakeholders and senior leadership, building cross-functional communication skills critical in enterprise

EDUCATION

M.S. Computer Science — Machine Learning Specialization | *Stevens Institute of Technology* *Expected Dec 2027*
GPA: 3.75/4.0
Relevant Coursework: Natural Language Processing, Big Data Technologies, Text Mining & Information Retrieval, Machine Learning, Deep Learning, Statistics, Probability Theory, Linear Algebra, Artificial Intelligence

B.S. Chemical Engineering | *Rutgers University* *Graduated May 2025*
Coursework: Process Design, Thermodynamics, Mass & Energy Transfer, Calculus, Differential Equations, Material Science

Awards: Cum Laude, Dean's List

LEADERSHIP & ACTIVITIES

Vice President | *ACOBA – East Coast, USA* *Jan 2025 - Present*

- Manage chapter finances and support executive decision-making as Vice President
- Co-organized the largest East Coast cricket festival, overseeing logistics, community partnerships, and charity fundraising demonstrating project management and stakeholder coordination at scale

Member, AIChE Chem-E-Car Team | *Rutgers University* *2024 - 2025*

- Participated in designing, building, and competing with chemically powered model cars in national AIChE competitions, applying engineering problem-solving in a high-stakes team environment

Member, Phi Theta Kappa Honor Society |*Academic Excellence Recognition* *2020 - 2022*

- Inducted for academic excellence and demonstrated leadership; PTK recognizes the top scholars across two-year institutions nationally